Why the SK order parameter has no gaps

A variational argument from contact to strict curvature

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The Sherrington–Kirkpatrick (SK) model asks us to choose signs that maximise a random sum of pair interactions. The Parisi formula expresses the limiting maximum through a nondecreasing function, the **order parameter**. We explain why a minimising order parameter cannot be constant on any interval below one: its increment measure has no gaps.

The proof compares two consequences of a hypothetical gap. Minimality fixes the values and bounds the slopes of an auxiliary curve at its endpoints. An analytic estimate makes the derivative strictly convex between them, leaving too little area to connect the required values. Initial and terminal gaps need separate arguments. We then return to the spins: a theorem about their energy landscape gives nearly optimal configurations at every prescribed approximate overlap.

We follow the quantitative proof in OpenAI (2026a); Hong-Bin Chen (2026), Theorem 1.3 had already proved full support by a different method. The estimate used here makes the final contradiction a direct comparison of areas. We assume calculus, integration against measures and expectation. With that background, the reader can reconstruct the no-gap argument from the analytic inputs stated at their uses. The stochastic calculation in Section 3.3 and the explanation of strict curvature in Section 7 are optional; the full analytic proofs remain in the cited sources.

1. The finite optimisation problem

At each of n positions, or **sites**, we choose a sign $\sigma_i \in \{-1, 1\}$, called a **spin**. A configuration is the vector $\sigma = (\sigma_1, \dots, \sigma_n)$. For each unordered pair choose an independent standard Gaussian J_{ij}, a normal random variable with mean zero and variance one. These coefficients, or **couplings**, are drawn once and then held fixed while we maximise the energy, or **Hamiltonian**,

$$H_n(\sigma) = \frac{1}{\sqrt{n}} \sum_{1 \le i < j \le n} J_{ij} \sigma_i \sigma_j.$$

A positive coupling rewards equal signs; a negative one rewards opposite signs. For example, positive couplings on pairs 12 and 23 favour $\sigma_1 = \sigma_2 = \sigma_3$, which conflicts with a negative coupling on pair 13. This inability to maximise every pair's contribution simultaneously is called **frustration**. The factor $n^{-1/2}$ fixes the model's scale.

For two spins, $H_2(\sigma) = J_{12}\sigma_1\sigma_2/\sqrt{2}$. Choosing equal or opposite signs according to the sign of J_{12} gives $\max_\sigma H_2(\sigma) = |J_{12}|/\sqrt{2}$. For many spins, we seek the limiting expected maximum per spin,

$$P_* := \lim_{n \to \infty} \frac{1}{n} \mathbb{E} \max_\sigma H_n(\sigma),$$

where expectation averages over the couplings after taking the maximum. **Zero temperature**, or **ground-state value**, refers here to this maximisation problem. The model is **pure** because every term contains exactly two spins, and has **zero external field** because there is no additional term $h \sum_i \sigma_i$.

The maximum is one question; the arrangement of nearly optimal choices is another. To compare two configurations for the *same couplings*, define their **overlap** $R(\sigma,\tau) = n^{-1}\sum_i \sigma_i\tau_i$, their average agreement with a disagreement counted as -1. If they disagree at k positions, then $R = 1 - 2k/n$. For example, $(+,+,+,+)$ and $(+,+,+,-)$ have overlap $1/2$; two disagreements would give zero. We will ask whether two nearly optimal configurations can differ on roughly one quarter of their sites, and more generally at any prescribed fraction up to one half.

Overlap is also the quantity that determines the energy covariance. For fixed σ, τ, put $v_i = \sigma_i\tau_i$. Independence of the couplings and $v_i^2 = 1$ give

$$\mathbb{E}[H_n(\sigma)H_n(\tau)] = \frac{1}{n}\sum_{i<j} v_i v_j = \frac{(\sum_i v_i)^2 - n}{2n} = \frac{n}{2}R(\sigma,\tau)^2 - \frac{1}{2}.$$

Dividing by n gives the limiting covariance $\xi(t) = t^2/2$. For exact covariance $n\xi(R)$, add the same independent centred Gaussian of variance $1/2$ to every energy; this leaves the expected maximum unchanged. Reversing one configuration's signs preserves its energy and reverses its overlap with another. We next define deterministic candidate values indexed by nonnegative overlap scales t. The Parisi theorem in Section 3 identifies their minimum with the limiting spin optimum; that connection is an external input, not a change of variables in the finite maximisation.

2. What a gap means

An admissible **order parameter** is a function $\gamma : [0,1) \to [0,\infty)$ that is nondecreasing, right-continuous and integrable. It assigns a cumulative weight to each overlap scale t; these weights will enter the variational formula. They are not probabilities of overlaps sampled from a finite system. The associated **Stieltjes measure** μ_γ records where the weight increases:

$$\mu_\gamma([0,t]) = \gamma(t), \qquad \mu_\gamma((a,b]) = \gamma(b) - \gamma(a).$$

The first equation allows mass $\gamma(0)$ at zero. Positive mass concentrated at one point is called an **atom**. The measure is finite on every $[0,T]$ with $T < 1$, although its total mass may be infinite. It is not required to be a probability measure. A unit of mass at s contributes to $\gamma(t)$ at every $t \geq s$. Integrating that contribution over t gives

$$\int_0^1 \gamma(t)\,dt = \int_{[0,1)} \left(\int_s^1 dt \right) \mu_\gamma(ds) = \int_{[0,1)} (1-s)\,\mu_\gamma(ds) < \infty.$$

The **support** consists of points every neighbourhood of which has positive measure. Neighbourhoods are taken within $[0,1)$: the endpoint one is not part of the domain. Full support therefore means that every nonempty relative open interval has positive mass. It is equivalent to strict increase of γ. Indeed, full support gives $\mu_\gamma((a,b]) > 0$ whenever $a < b$; conversely, strict increase gives positive mass on a smaller interval inside every neighbourhood.

For example, the admissible candidate

$$\gamma(t) = \begin{cases} t, & 0 \le t \le 1/3, \\ 1/3, & 1/3 \le t \le 2/3, \\ t - 1/3, & 2/3 \le t < 1 \end{cases}$$

has a gap on $(1/3, 2/3)$. Its measure has density one outside that interval and zero inside it: the measure of a set is the integral of this density over the set. Both endpoints of the gap remain in the support. The upper curve in Figure 1 stays positive across the gap; it is its *increments* that vanish. This candidate has infinitely many support points, including points arbitrarily close to one, yet still has a middle gap. Those two weaker properties do not give full support.

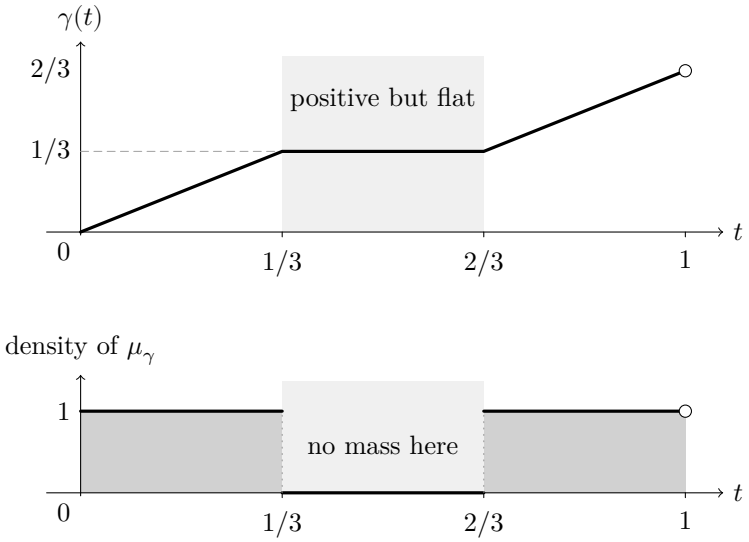


Figure 1: The same candidate in two representations. The upper curve is the piecewise-linear γ from Section 2; the lower plot shows the density of its Stieltjes measure. On $(1/3, 2/3)$ the cumulative value stays positive while its increments vanish. The endpoints $1/3$ and $2/3$ still belong to the support: every neighbourhood of either contains positive mass. The lower density belongs to this example; a general Stieltjes measure need not have one. Open circles mark the excluded endpoint 1. The candidate is not a computed minimiser. Axes and guides adapt OpenAI (2026a), Figure 1; the paired example is constructed here.

A constant candidate $\gamma(t) = c > 0$ makes the distinction sharper: its measure is $c\delta_0$, supported only at zero. Positivity of the function is not the conclusion we seek. Nor does full support alone assert a density: a measure with summable positive masses at a dense countable set has full support. The further-reading note returns to the stronger density conclusion.

3. How a candidate receives its value

3.1 An auxiliary optimisation

To assign a value to γ, we first solve a one-dimensional control problem: bias a noisy path so that its final position is large in absolute value, while paying a running cost for the bias. Varying γ will then reveal where a minimiser can have mass. Here the overlap scale t plays the role of time; the path is auxiliary, not a motion of the spins.

The auxiliary noise is Brownian motion B: a continuous random path starting at zero, with independent centred Gaussian increments, where $B_s - B_t$ has variance $s - t$. Choose a **control** $\alpha_s \in [-1, 1]$: at each time s, this rule chooses a response using only the Brownian increments observed since the starting time t. The rule must be measurable and must not use future noise; the formal requirement is progressive measurability. For a starting position x, set

$$\Phi_\gamma(t, x) = \sup_\alpha \mathbb{E}\left[\left|x + B_1 - B_t + \int_t^1 \gamma(s)\alpha_s\, ds\right| - \frac{1}{2}\int_t^1 \gamma(s)\alpha_s^2\, ds\right].$$

The function $\Phi_\gamma(t, x)$ gives the best expected reward, less the running cost, when we start at time t in state x. Inside the absolute value, noise and control change that state before the final reward is paid. The reward is a one-spin optimisation, $|x| = \max_{\sigma\in\{-1,1\}} x\sigma$. Expectation here averages over Brownian paths, with γ fixed. The **Parisi functional** assigns the candidate its value:

$$\mathcal{P}(\gamma) = \Phi_\gamma(0, 0) - \frac{1}{2}\int_0^1 t\gamma(t)\, dt.$$

The inner supremum chooses a response for fixed γ; the outer problem minimises $\mathcal{P}$ over γ. The ground-state Parisi formula and attainment give

$$P_* = \min_{\gamma \text{ admissible}} \mathcal{P}(\gamma).$$

Every candidate therefore supplies an upper bound; minimisation finds the sharpest one. The variational equality and attainment come from Auffinger–Chen (2017), Theorem 1 and the proof of Lemma 3; Gaussian concentration also gives the expected-value form used here. The no-gap theorem assumes a minimiser and proves a property of it; it does not prove existence. For comparison, $\gamma = 0$ gives $\Phi_0(t, x) = \mathbb{E}|x + \sqrt{1 - t}Z|$ and $\mathcal{P}(0) = \sqrt{2/\pi}$ for a standard Gaussian Z. Evaluating this candidate alone does not establish minimality.

3.2 The best response and the path it generates

At $t = 1$ there is no noise or response left to accumulate, so the value is $|x|$. At earlier times it includes the best remaining response. Its partial differential equation (PDE) runs backward from that final value:

$$\partial_t\Phi_\gamma = -\frac{1}{2}\left(\partial_{xx}\Phi_\gamma + \gamma(t)(\partial_x\Phi_\gamma)^2\right), \qquad \Phi_\gamma(1, x) = |x|.$$

Write $u = \partial_x\Phi_\gamma$ for its spatial slope and $a = \partial_{xx}\Phi_\gamma$ for its spatial curvature. The source proves convexity and $|u| \le 1$, hence $a \ge 0$. The equation expresses a recursive choice: choose a response for a short first step, then use the best continuation from the state reached. At a continuity point of γ, this gives the formal first-order relation

$$\Phi_\gamma(t, x) = \sup_{|v|\le 1}\left\{\mathbb{E}\Phi_\gamma(t + h, x + \gamma(t)vh + \sqrt{h}Z) - \tfrac{1}{2}\gamma(t)v^2h\right\} + o(h).$$

Subtract the present value, divide by h, and Taylor-expand. The identities $\mathbb{E}Z = 0$ and $\mathbb{E}Z^2 = 1$ give

$$0 = \Phi_t + \tfrac{1}{2}\Phi_{xx} + \sup_{|v|\le 1} \gamma(t)(v\Phi_x - v^2/2).$$

At slope u, completing a square shows which response to choose:

$$\gamma v u - \tfrac{1}{2}\gamma v^2 = \tfrac{1}{2}\gamma u^2 - \tfrac{1}{2}\gamma(v-u)^2.$$

Since $\gamma \ge 0$ and $|u| \le 1$, $v = u$ attains the supremum, yielding the stated PDE. This calculation explains the equation; justifying the recursion and the Taylor expansion requires the control representation and regularity results. See Auffinger–Chen, Theorem 2 for the zero-temperature control formula and Jagannath–Tobasco, Lemma 18 for the control method with finite-temperature terminal data. The extension to the coefficients used here is in OpenAI (2026a), Lemmas 2.1–2.2.

Under the slope response the position follows

$$X_t = B_t + \int_0^t \gamma(s)u(s, X_s)\, ds, \qquad X_0 = 0.$$

This random path is the **associated diffusion**. The response is a **feedback control**: at each time it uses the slope at the current state X_t. An atom of mass c at scale s gives $\gamma(t) = c\mathbf{1}_{[s,1)}(t)$: it switches on both the drift and the running cost from that scale onward. It does not make X jump, because the accumulated drift is a time integral. The path's average squared slope response defines the deterministic curve

$$q(t) := \mathbb{E}[u(t, X_t)^2].$$

The first variation will show that $\frac{1}{2}(q(t) - t)$ is the first-order cost density for changes in γ. This makes q useful: optimality first forces $q(t) = t$ where the minimiser has mass; excluding gaps then extends that equality everywhere.

3.3 How the squared response grows

For any admissible candidate, the facts needed below are

$$q \in C^1([0,1)), \qquad q(0) = 0, \qquad 0 \le q(t) \le 1, \qquad q'(t) = \mathbb{E}[a(t, X_t)^2].$$

Thus spatial curvature controls growth of the squared response. Derivatives at zero are from the right, and q' is continuous even across jumps of γ. These facts do not assert $q = t$ for a general candidate. They follow from OpenAI (2026a), Lemmas 2.1–2.2; readers who use them as inputs can continue to Section 4. The following calculation explains the moment identity.

Itô's rule is the chain rule for Brownian paths. Along $dX_t = b_t\, dt + dB_t$, the coefficient of dt (the **drift**) in a smooth $f(t, X_t)$ is $f_t + b_t f_x + f_{xx}/2$, and its noise term is $f_x\, dB_t$. Differentiate the PDE in x and take $b_t = \gamma u$. The drift of the slope cancels:

$$du(t, X_t) = \underbrace{(u_t + \gamma u u_x + u_{xx}/2)}_{0}\, dt + a(t, X_t)\, dB_t.$$

Thus $M_t = u(t, X_t)$ is a bounded **martingale**: its conditional expected later value, given observations up to time t, is its present value. Zero field gives symmetry and $M_0 = u(0, 0) = 0$. Applying Itô's rule once more to M_t^2 gives

$$d(M_t^2) = 2M_t a(t, X_t)\, dB_t + a(t, X_t)^2\, dt.$$

The noise term has mean zero because its coefficient is nonanticipating and bounded on each strip ending before one. Taking expectations and integrating yields $q(t) = \int_0^t \mathbb{E}[a(s, X_s)^2]\, ds$, which gives the asserted derivative. Since $\mathbb{E}M_t = 0$, $q(t)$ is also the variance of the slope response.

For each fixed $T < 1$, the required spatial derivatives are bounded on $[0, T] \times \mathbb{R}$, and step-coefficient and smooth-terminal approximations justify the identities. The bounds depend on the coefficient and strip. The PDE holds almost everywhere in time. These are not bounds up to time one.

4. The full-support theorem

Source theorem (OpenAI (2026a), Theorem 1.1). In the pure, zero-field model with $\xi(t) = t^2/2$, let γ be an admissible integrable minimiser of $\mathcal{P}$. Its Stieltjes measure has full relative support on $[0, 1)$, and γ is strictly increasing. For every associated diffusion and every $0 \le t < 1$,

$$\mathbb{E}[u(t, X_t)^2] = t, \qquad \mathbb{E}[a(t, X_t)^2] = 1.$$

The cumulative function determines its measure uniquely; for a fixed Brownian motion the associated diffusion is unique by the cited regularity results. The theorem requires a minimiser, not merely an admissible candidate. Zero field supplies $q(0) = 0$; $\xi'' = 1$ enters the curvature estimate. The statement does not cover nonzero field or general mixed-spin covariance functions.

5. Why a flat interval cannot minimise

5.1 Allowed changes and their first-order cost

The **first variation** in a direction η is the derivative at zero of $\varepsilon \mapsto \mathcal{P}(\gamma + \varepsilon\eta)$. At a minimum its right derivative is nonnegative whenever small $\varepsilon \ge 0$ leave the candidate admissible. It need not vanish: the opposite change may violate nonnegativity or monotonicity. Precisely, let $\eta \in L^1(0, 1)$ and suppose $\gamma + \varepsilon\eta$ stays admissible for every $0 \le \varepsilon \le \varepsilon_0$, for some $\varepsilon_0 > 0$. OpenAI (2026a), Lemma 6.1 proves

$$D\mathcal{P}(\gamma)[\eta] = \frac{1}{2} \int_0^1 \eta(t)(q(t) - t)\, dt.$$

Here q belongs to the unperturbed candidate γ. The term $-t/2$ comes from the explicit penalty in $\mathcal{P}$; $q(t)/2$ is the sensitivity of the reoptimised control value. The latter is the analytic content of the formula, not differentiation with a fixed control. The proof compares control values and

bounds the terminal remainder by the L^1 tail of η; related zero-temperature variation formulas are in Chen–Handschy–Lerman (2018), Propositions 2–4.

A candidate that fails the test. Return to $\gamma = 0$. Every control then gives the same value, since both its drift and its cost vanish. The slope response is still well defined. With F_{G} the standard normal distribution function, the heat formula gives

$$X_t = B_t, \qquad u_0(t,x) = 2F_{\mathrm{G}}\left(\frac{x}{\sqrt{1-t}}\right) - 1, \qquad a_0(t,x) = \sqrt{\frac{2}{\pi(1-t)}}e^{-x^2/(2(1-t))}.$$

The formula for u_0 is the expected sign of $x + \sqrt{1-t}Z$: it subtracts the probability of a negative final state from that of a positive one. Here u_0, not an arbitrarily chosen optimal control, defines q_0. Using $\mathbb{E}e^{-B_t^2/(1-t)} = \sqrt{(1-t)/(1+t)}$ in the moment identity and then integrating from zero yields

$$q_0'(t) = \frac{2}{\pi\sqrt{1-t^2}}, \qquad q_0(t) = \frac{2}{\pi}\arcsin t.$$

At the fixed state zero, symmetry gives $u_0(1/2, 0) = 0$. But q_0 averages the square at the random state $B_{1/2}$, and $q_0(1/2) = 1/3$. Equality $q = t$ is not a definition. The constant direction $\eta = 1$ is admissible and adds an atom at zero. Its cost is

$$D\mathcal{P}(0)[1] = \frac{1}{2}\int_0^1 \left(\frac{2}{\pi}\arcsin t - t\right) dt = \frac{1}{4} - \frac{1}{\pi} < 0.$$

So a small positive constant coefficient improves the zero candidate. This calculation uses the heat and variation formulas, not a minimiser hypothesis; see also Chen–Handschy–Lerman, proof of Theorem 6.

Adding a small atom at scale s raises the cumulative function at *every* later scale. It is always allowed: the total mass is not fixed, so no compensating removal is required. Its direction is $\eta(t) = \mathbf{1}_{[s,1)}(t)$. Thus

$$D\mathcal{P}(\gamma)[\mathbf{1}_{[s,1)}] = \frac{1}{2}\int_s^1 (q(t) - t)\, dt =: \frac{1}{2}G(s).$$

For the candidate in Figure 1, adding ε mass at $s = 1/2$ creates a jump of height ε there and raises the entire remaining curve by ε. This explains the tail integral: $G(s)/2$ is the marginal cost of placing mass at s. At a minimiser, $G(s) \geq 0$ for every $s < 1$.

We can also remove a fraction of existing mass on $K = [0, T]$, $T < 1$. The perturbed measure $(1 - \varepsilon\mathbf{1}_K)\mu_\gamma$ is nonnegative for $0 \leq \varepsilon \leq 1$; its cumulative function is $\gamma(t) - \varepsilon\mu_\gamma(K \cap [0,t])$. Thus the direction $\eta_K(t) = -\mu_\gamma(K \cap [0,t])$ is admissible. For Figure 1, take $K = [0, 1/3]$. Then $\eta_K(t) = -\min(t, 1/3)$: the perturbed curve has slopes $1 - \varepsilon$, 0, 1 on its three pieces, and plateau height $(1 - \varepsilon)/3$. It remains nonnegative and nondecreasing for $0 \leq \varepsilon \leq 1$. This example verifies admissibility; it does not assume that the pictured candidate minimises the functional.

Return now to a minimiser. Local finiteness makes η_K bounded. As in the weighted-mass calculation in Section 2, interchanging the integrals converts a cumulative change into a cost per unit of mass; $|q(t) - t| \le 1$ justifies the interchange. We obtain

$$0 \le D\mathcal{P}(\gamma)[\eta_K] = -\frac{1}{2}\int_K G(s)\,\mu_\gamma(ds).$$

Since $G \ge 0$, this integral must be zero. If G were positive at a support point, continuity would make it positive throughout a neighbourhood of positive measure, contradicting this equality for a large enough $T < 1$. Therefore $G = 0$ on the support. This equality is called **contact**: the nonnegative function touches zero there. Removing a fraction of existing mass makes the argument valid for continuous measures too: subtracting an atom where none exists would not be an admissible change.

5.2 Contact gives endpoint conditions

We need information about q at the ends of a possible gap. An interior support point s is a minimum of G. Since $G'(s) = s - q(s)$ and $G''(s) = 1 - q'(s)$, ordinary calculus gives

$$q(s) = s, \qquad q'(s) \le 1.$$

At zero, if it belongs to the support, contact and symmetry give $G(0) = q(0) = 0$. The expansion $G(h) = \frac{1}{2}(1-q'(0))h^2 + o(h^2)$ and nonnegativity imply $q'(0) \le 1$. This one-sided argument includes an atom at zero.

5.3 Strict curvature creates an impossible integral

The support is relatively closed in $[0, 1)$. Every missing point lies in a maximal interval of its complement. Such an interval has two support endpoints, starts at zero outside the support, or reaches one. We treat these three cases in order; the last includes empty support.

First suppose (r, s) has both endpoints in the support, with $0 \le r < s < 1$. Its constant height c is necessarily positive: for t inside the gap, a small relative neighbourhood of r lies in $[0, t]$ and has positive mass, so $c = \mu_\gamma([0, t]) > 0$. If $r = 0$, this mass is an atom at zero. Contact imposes

$$q(r) = r, \quad q(s) = s, \quad q'(r) \le 1, \quad q'(s) \le 1.$$

The first two conditions require the average of q' across the gap to be one. The next input will force it below one throughout.

Strict-curvature input (OpenAI (2026a), Proposition 5.1). For any admissible coefficient equal to $c > 0$ on an open interval, its moment curve satisfies, strictly inside that interval,

$$q'''(t) \ge \frac{c^2}{1000}\mathbb{E}[a(t, X_t)^4] > 0 \qquad (r < t < s).$$

Section 7 explains how this sign is obtained after expectation. The final inequality uses $a(t, x) > 0$, proved by Gaussian smoothing on the constant interval (OpenAI (2026a), Lemma 2.3); convexity of Φ alone would only give $a \ge 0$. Because $q''' = (q')''$, the function q' is strictly convex: between two

points its graph lies strictly below their joining chord. The moment identity makes q' continuous at the endpoints, even if γ jumps there; higher derivatives are needed only inside the gap. Hence

$$q'(t) < \frac{s-t}{s-r}q'(r) + \frac{t-r}{s-r}q'(s) \le 1 \qquad (r < t < s).$$

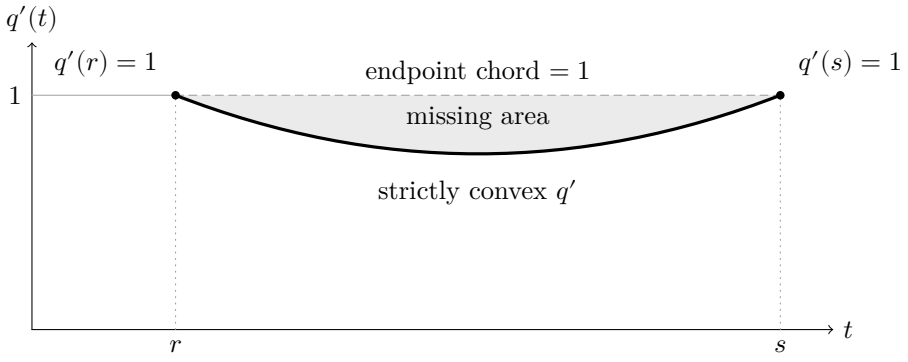


Figure 2: The case where strictness is indispensable: both endpoint slopes equal one, so the chord alone loses no area. A strictly convex slope lies below it. Contact would require zero shaded area, but $\int_r^s (1 - q'(t))\,dt > 0$. The drawn calculus model is $1 - x(1-x)$ with $x = (t-r)/(s-r)$; its mean is $5/6$, while the constant slope one has mean one. Neither is asserted to be an SK moment curve. The proof also allows unequal endpoint slopes, each at most one. Adapted from Figure 1 of OpenAI (2026a), OpenAI/math (Apache 2.0).

Integration gives $q(s) - q(r) < s - r$, contradicting contact. For the candidate in Section 2, a minimising plateau on $(1/3, 2/3)$ would require the same increment to be both equal to and strictly less than $1/3$.

5.4 The remaining initial and terminal gaps

Initial gap with zero outside the support. If $[0, s)$ is missing and $s < 1$ is the first support point, then $\gamma = 0$ on this interval. The positive-curvature bound vanishes. Instead the heat equation gives $q''(t) = \mathbb{E}[\Phi_{xxx}(t, X_t)^2] > 0$ for $0 < t < s$, so q' is strictly increasing. The strict inequality is only at positive times: at zero, symmetry gives $\Phi_{xxx}(0,0) = 0$. Here $X_t = B_t$, but the heat equation evolves backward from $\Phi_\gamma(s, \cdot)$, which still depends on the later coefficient. The explicit arcsine formula for the globally zero candidate therefore does not apply. For $0 < t < s$, B_t has a positive density, and $\Phi_{xxx}(t, \cdot)$ cannot vanish identically: then the even function $\Phi(t, \cdot)$ would be quadratic. Bounded slopes would make it constant. But the zero control and Jensen's inequality give $\Phi_\gamma(t, x) \ge \mathbb{E}|x + B_1 - B_t| \ge |x|$, excluding constancy. Thus $q'(t) < q'(s) \le 1$. Symmetry supplies $q(0) = 0$ without contact at zero, while right contact gives $q(s) = s$. Integration again gives too little area. The case with zero in the support was already included in Section 5.3, using its atom and the one-sided slope bound.

Terminal gaps. Figure 2 requires contact at both ends. It cannot handle a gap reaching one, which lies outside the domain. Even the zero candidate has $q_0(t) \to 1$ while $q_0'(t) \to \infty$; convergence of the value supplies no endpoint slope bound. Instead OpenAI (2026a), Lemma 6.3 proves the following terminal estimate: if $\gamma = c$ on $(r, 1)$ for a finite $c \ge 0$, then for some $C > 0$ and all sufficiently late t,

$$1 - q(t) \geq C\sqrt{1-t}.$$

The two bounds behind this estimate have different jobs. Fix an earlier time inside the assumed plateau and a bounded interval containing positive probability mass of the diffusion then. The source's explicit transition kernel carries some of that mass to a neighbourhood of zero with a density bounded below uniformly for late t. The elapsed time from the fixed earlier time stays positive; this is not an estimate obtained by extending compact-strip derivative bounds to one.

The second source bound is $1 - u(t,x)^2 \geq \kappa > 0$ throughout $|x| \leq \sqrt{h}$, where $h = 1 - t$ and t is sufficiently late. On a constant terminal interval, the heat formula writes u as a positively weighted average of the final signs $+1$ and -1. The source bounds both weights away from zero in this window, whose width matches the remaining noise scale $\sqrt{h}$. If $m > 0$ is the density lower bound there, these two estimates give

$$1 - q(t) \geq \int_{-\sqrt{h}}^{\sqrt{h}} \kappa m \, dx = 2m\kappa\sqrt{h}.$$

Both uniform bounds are source inputs under the terminal-plateau hypothesis. The elementary integration explains the square root, which dominates the smaller quantity h near zero.

It follows that $q(t) - t \leq (1-t) - C\sqrt{1-t} < 0$ near one, and

$$G(s) \leq \frac{(1-s)^2}{2} - \frac{2C}{3}(1-s)^{3/2} < 0$$

for s sufficiently close to one. This contradicts the nonnegativity forced by adding mass. It also excludes the empty-support candidate $\gamma = 0$. The explicit late-step variation against that candidate appears in Chen–Handschy–Lerman, proof of Theorem 6. Auffinger–Chen–Zeng, Remark 7 had also observed that a minimiser cannot have a terminal plateau; the estimate used here is OpenAI's Lemma 6.3.

These cases exhaust the complement of the support. Contact now gives $q(t) = t$ for every interior time, and symmetry includes zero. Differentiating gives $\mathbb{E}[a(t, X_t)^2] = 1$, completing the theorem.

6. Return to the spins

We can now return to the arrangement of nearly optimal spin choices. A separate theorem of Auffinger–Chen–Zeng, Proposition 1, equation (5) applies at each point in the minimiser's support. Combining it with full support: for every fixed overlap $r \in [0,1)$ and tolerances $\varepsilon, \eta > 0$, with probability over the couplings tending to one exponentially fast as $n \to \infty$, there exist configurations σ, τ with

$$|R(\sigma,\tau) - r| < \varepsilon, \qquad H_n(\sigma)/n,\ H_n(\tau)/n \geq P_* - \eta.$$

If k sites differ, $k/n = (1-R)/2$. The overlap tolerance therefore gives

$$\left|\frac{k}{n} - \frac{1-r}{2}\right| < \frac{\varepsilon}{2}.$$

At target overlap 1/2, two nearly optimal choices can differ on approximately one quarter of the sites; at target zero, on half. The pair may depend on the chosen target and the couplings. This is existence for each fixed target, not a statement about typically sampled pairs or a procedure for finding them. The companion *Long Reasoning Record*, §9.3, "Near-ground configurations at a prescribed overlap", gives the probability bound and the covariance adjustment.

A check on the area argument

Question. Suppose a differentiable function f on $[2,5]$ has $f(2) = 2$, $f(5) = 5$, $f'(2) = 1/2$ and $f'(5) = 3/2$. If f' is strictly convex, can these data hold? What changes if strict convexity is weakened to convexity? The endpoint chord is now tilted, so a pointwise comparison with the constant one is unavailable.

Try the comparison before reading the answer below.

Answer. A strictly convex f' lies below its endpoint chord. That chord has mean height $(1/2 + 3/2)/2 = 1$, hence integral three. Therefore $f(5) - f(2) = \int_2^5 f'(t)\,dt < 3$, contradicting the prescribed values. With non-strict convexity the data are possible: $f(t) = 2 + (t-2)/2 + (t-2)^2/6$ has affine derivative and satisfies all four endpoint values. The argument needs a strict loss of area; the common endpoint upper bound of one in Figure 2 is convenient, not essential. This is a calculus example, not a candidate SK moment function.

7. Where the strict-curvature input comes from

The proof in Section 5 used strict curvature on a positive constant interval. This section explains how that expectation acquires a sign; the full analytic proof remains in the cited sources. For this section, take any admissible coefficient with $\gamma = c > 0$ on an open interval, without assuming minimality. Set $b = \Phi_{xxx}$ and $d = \Phi_{xxxx}$ there. Differentiating the PDE and applying Itô's rule gives

$$q'' = \mathbb{E}[b^2 - 2ca^3], \qquad q''' = \mathbb{E}[d^2 - 12cab^2 + 6c^2a^4].$$

The negative middle term prevents a pointwise positivity argument. The source instead uses the spatial probability density $f_t(x)$ of X_t. Its probability law $f_t(x)\,dx$ is a measure in x, distinct from the Stieltjes measure $\mu_\gamma(ds)$ on overlap scales. For step coefficients and smoothed terminal data, this density factors into positive functions propagated from the two ends of the time interval. Bounds on their logarithmic derivatives supply the signs needed in integration by parts.

The operation can be written directly for the positive density f_t. For a smooth function F with integrable terms and vanishing boundary product Ff_t,

$$\mathbb{E}\left[F'(X_t) + F(X_t)\frac{f_t'(X_t)}{f_t(X_t)}\right] = \int_{\mathbb{R}} (Ff_t)'(x)\,dx = 0.$$

For a small example of this operation, let Z be standard normal and $F(x) = 2x$. Its density has logarithmic derivative $-x$, so $\mathbb{E}[2 - 2Z^2] = 0$. Although $2x^2 - 1$ is negative near zero, adding that zero-mean correction gives

$$(2x^2 - 1) + (2 - 2x^2) = 1.$$

Hence $\mathbb{E}[2Z^2 - 1] = 1 > 0$. The boundary term vanishes because a polynomial times the Gaussian density tends to zero. This illustrates how a correction changes the visible sign without changing the expectation; it supplies none of the SK-specific sign estimates.

At the step-coefficient, smooth-terminal stage, write $R_c = d^2 - 12cab^2 + 6c^2a^4$. The source constructs a correction C from the spatial derivatives and the density factors, with

$$\mathbb{E}[C(t, X_t)] = 0, \qquad R_c(t, x) + C(t, x) \geq \frac{c^2}{1000} a(t, x)^4.$$

A rational polynomial identity gives the second inequality by writing the remainder as a sum of nonnegative terms. The density bounds and vanishing boundary products justify the first. Consequently

$$q'''(t) = \mathbb{E}[R_c(t, X_t)] = \mathbb{E}[(R_c + C)(t, X_t)] \geq \frac{c^2}{1000} \mathbb{E}[a(t, X_t)^4].$$

This is how the correction produces a useful lower bound for the expectation even though R_c can be negative at particular states.

To reach an arbitrary admissible coefficient, fix a time inside the plateau and approximate in L^1 by step coefficients, keeping the same value c near that time. Uniform convergence of spatial derivatives through order four on a fixed strip below one, together with convergence of the coupled paths, passes the quantitative inequality to the limit. Strictness is then established for the limit itself: the constant-coefficient heat formula expresses a as a nonnegative mean plus c times a strictly positive variance of a nonconstant slope under a positively weighted Gaussian law. Hence $a(t, x) > 0$ and its fourth moment is positive. Positivity of the approximants alone would not suffice. The auxiliary logarithmic derivatives need not converge: they have already been removed by integration.

The analytic details are OpenAI (2026a), §§3–5 and the companion *Long Reasoning Record*, §4.4, "How the signs make a correction with zero expectation useful", and §4.5, "A fixed-time inequality and its plateau application".

Further reading

Full support alone does not give a density. Additional differentiation of the identities $q = t$ and $\mathbb{E}[a^2] = 1$, followed by another use of the integrated curvature estimate on locally constant comparison coefficients, proves that γ is smooth, $\gamma(0) = 0$, and $\mu_\gamma = \gamma'(t)\, dt$ with $\gamma'(t) > 0$ for $0 < t < 1$ (OpenAI (2026a), Corollary 6.4). The *Long Reasoning Record*, §§9.1–9.2, supplies this additional argument; Chen's earlier theorem had already given smooth nonnegative density and full support.

Further martingale arguments recover the ground-state value from the terminal slope response. They use the first curvature moment $\mathbb{E}[a]$, distinct from the second moment $\mathbb{E}[a^2] = 1$ used here. See OpenAI (2026a), Corollary 7.1 and the *Long Reasoning Record*, §6.

The separate circuit theorem in OpenAI (2026b) resolves the eventual-optimality conjecture for the Quantum Approximate Optimization Algorithm (QAOA) in Basso–Farhi–Marwaha–Villalonga–Zhou, §7, equation (7.2). The *Long Reasoning Record*, §8, develops its circuit construction and the order of limits. It requires additional arguments beyond full support.

Sources and scope of verification

This account reconstructs selected parts of OpenAI's analytic manuscript. The Lean implementation, challenge statement and Comparator configuration concern a minimiser's support and moment identities. They assume a minimiser; they do not prove the external Parisi formula or attainment. The separate scalar implementation and statement concern the value identity and finite Gaussian approximation. Neither selected statement covers positive density, the external near-ground overlap theorem used in Section 6, or the complete QAOA circuit theorem. The implementations use Lean 4 (de Moura–Ullrich, 2021) and mathlib (The mathlib Community, 2020).

Comparator acceptance and research-v2 extraction remain unknown; see the *Long Reasoning Record*, §13. Kernel checking establishes a declared statement under its definitions and axioms. Its correspondence with the analytic theorem, independent proof review and reader comprehension are separate questions. The links below select the editions used.

- **OpenAI (2026a).** *Full support of the zero-temperature Sherrington–Kirkpatrick order parameter*, 27 September 2026. Pinned manuscript. §2 supplies regularity, Proposition 5.1 the strict curvature estimate, §6 the support proof, and Corollary 6.4 the additional density conclusion.
- **Antonio Auffinger and Wei-Kuo Chen (2017).** *Parisi formula for the ground state energy in the mixed p-spin model*, Annals of Probability 45(6B), 4617–4631. Version 2. The variational formula and attainment used in Section 3.1.
- **Aukosh Jagannath and Ian Tobasco (2016).** *A dynamic programming approach to the Parisi functional*, Proceedings of the American Mathematical Society 144, 3135–3150. Version 4. The control method, with finite-temperature terminal data; it is not by itself the zero-temperature regularity theorem used here.
- **Wei-Kuo Chen, Madeline Handschy and Gilad Lerman (2018).** *On the energy landscape of the mixed even p-spin model*, Probability Theory and Related Fields 171, 53–95. Version 4. Zero-temperature analytic and variational results and the explicit zero-candidate descent.
- **Antonio Auffinger, Wei-Kuo Chen and Qiang Zeng (2020).** *The SK model is infinite step replica symmetry breaking at zero temperature*. Version 2, Proposition 1 and Remark 7. Near-ground overlap pairs and the earlier terminal-plateau observation.
- **Hong-Bin Chen (2026).** *FRSB in the SK spin glass: convergence to full-interval support at zero temperature*. Version 1, Theorem 1.3. Full support and smooth nonnegative density; the strict positivity cited in the further-reading note comes from the separate OpenAI corollary.
- **OpenAI (2026b).** *QAOA attains the SK ground-state energy in the thermodynamic-first limit*, 25 September 2026. Pinned manuscript. The separate circuit theorem.
- **Joao Basso, Edward Farhi, Kunal Marwaha, Benjamin Villalonga and Leo Zhou (2022).** *The Quantum Approximate Optimization Algorithm at High Depth for MaxCut on Large-Girth Regular Graphs and the Sherrington-Kirkpatrick Model*. Full preprint, version 3, §7, equation (7.2). The eventual-optimality conjecture.